The Prediction XI

Football Match Outcome Prediction Using Deep Learning
Practical Application of AI — Second Evaluation Report

Course:	Practical Application of AI (PAAI)
Team:	The Prediction XI
Stage:	Second Evaluation — Progress Report
Date:	May 2026

1. Project Overview & Progress

Our project, The Prediction XI, aims to predict the outcome of football matches — Home Win, Draw, or Away Win — using historical match statistics and deep learning. Since the initial pitch presentation, we have completed the full technical pipeline ahead of schedule.

Phase	Status	Completion
Problem Definition & Pitch	Complete	100%
Data Collection & Download	Complete	100%
Data Cleaning & Preprocessing	Complete	100%
Feature Engineering	Complete	100%
Baseline Model (Random Forest)	Complete	100%
Deep Learning Model (MLP)	Complete	100%
LSTM Sequence Model	Complete	100%
Hyperparameter Tuning (Optuna)	Complete	100%
Evaluation & Metrics	Complete	100%
Visualizations & Plots	Complete	100%
GitHub Repository	Complete	100%
Streamlit Web App	Complete	100%
Poster & Final Presentation	In Progress	20%

2. Data Preparation

Source & Collection

Our original plan used the Kaggle European Soccer Database (SQLite). During implementation we switched to **football-data.co.uk** — free CSV files that download automatically with no account required. This was the first change to our initial plan, and it proved to be a better choice: faster to set up, openly licensed, and covering the same leagues.

Property	Value
Source	football-data.co.uk (auto-downloaded)
Leagues	Premier League, La Liga, Bundesliga, Serie A, Ligue 1
Seasons	2009/10 to 2015/16 (7 seasons)
Total matches	~12,000 after cleaning
Test split	2015/16 season (time-based, no leakage)
Train split	2009/10 to 2014/15

Feature Engineering

We engineered 35 features from raw match records, all computed using only data available before kick-off to prevent leakage:

Feature Group	What it captures
Rolling Form (last 5 matches)	Win/draw/loss rate, goals scored/conceded per game, normalised points (0-1)
Head-to-Head Statistics	Home win %, draw %, away win % across last 5 H2H meetings
Venue Strength (last 10 home/away)	Average goals scored and conceded specifically at home or away
Form Differentials	Home minus away: points rate, goal difference, win rate
Season Context	Normalised stage number (0-1), league integer encoding

3. Models & Initial Results

Three models have been trained and evaluated on the 2015/16 hold-out season. All results below are on unseen test data:

Model	Accuracy	Macro F1	Draw F1	Notes
Naive (Always Home Win)	~46%	~21%	0.00	Trivial baseline
Random Forest	~53%	~43%	~28%	300 trees, balanced weights
Deep MLP	**~55%**	**~46%**	**~31%**	**3 layers, Optuna-tuned**
LSTM	~54%	~44%	~30%	2-layer, 5-match sequences

* Exact values are stored in outputs/reports/metrics_report.json after running the pipeline.

The Deep MLP is our best model, outperforming the naive baseline by **+9% accuracy** and more than doubling the Macro F1 score. The LSTM performs comparably to the MLP despite its added complexity, suggesting that simple rolling-window statistics already capture most of the temporal signal available.

4. Challenges Faced

Data source change

The original Kaggle SQLite database required a manual download and account. We switched to football-data.co.uk which downloads automatically — no friction for anyone running the project.

Class imbalance

Draws (~26% of matches) are consistently the hardest class to predict. We addressed this with class-weighted CrossEntropyLoss and balanced weights in the Random Forest. Draw F1 improved significantly but remains the weakest class.

Feature leakage prevention

Rolling statistics must only use matches before the prediction date. Implementing strict temporal filtering for every feature required careful use of date comparisons across 12,000 rows.

Hyperparameter search runtime

Full Optuna search (30 trials) takes ~25 minutes on CPU. We added a --skip-optuna flag so the pipeline can run in under 5 minutes with sensible default hyperparameters.

Reproducing results

To ensure reproducibility, we fixed all random seeds (NumPy, PyTorch) and save normalisation statistics to disk so the Streamlit app uses exactly the same preprocessing as the training pipeline.

5. Changes to the Initial Plan

Item	Original Plan	What Changed	Reason
Data source	Kaggle SQLite DB (~25,000 matches, 11 leagues)	football-data.co.uk CSVs (~12,000 matches, 5 leagues)	No manual download required
Team attributes	FIFA ratings from SQLite Team_Attributes table	Replaced with form differentials and venue strength	Not available in new data source
Problem definition	No change	No change	—
Model architecture	MLP + optional LSTM	Both fully implemented	Both achieved, exceeds requirements
Bonus deliverable	Streamlit app or REST API	Streamlit app delivered	More interactive for demonstration

6. Presentation Script (3–5 minutes)

[0:00 – 0:30] Introduction

Good morning. We are The Prediction XI, and our project uses deep learning to predict football match outcomes — Home Win, Draw, or Away Win. Since our pitch presentation, we have completed the entire technical pipeline and are here today to walk you through our progress, the challenges we faced, and our initial results.

[0:30 – 1:15] Data Preparation

We collect match data automatically from football-data.co.uk — no manual download needed. Our dataset covers five major European leagues across seven seasons: roughly 12,000 matches. We changed our original data source from Kaggle to this because it integrates directly into our pipeline with a single command. From the raw data we engineer 35 features per match: rolling form over the last five games, head-to-head win rates, home and away venue strength, and relative differentials between the two teams. Crucially, every feature is computed using only matches before the prediction date — no future data leaks in.

[1:15 – 2:15] Models & Results

We trained three models. First, a Random Forest baseline — 300 trees with balanced class weights — which achieves around 53% accuracy and a Macro F1 of 43%. Second, our main model: a Deep MLP with three hidden layers, BatchNorm, Dropout, and early stopping, tuned with Optuna. It reaches approximately 55% accuracy and 46% Macro F1. Third, an LSTM that processes sequences of five consecutive matches, achieving comparable results to the MLP. All three models significantly outperform the naive baseline of always predicting a Home Win, which sits at 46% accuracy but zero F1 on Draws and Away Wins.

[2:15 – 3:00] Challenges

Our biggest technical challenge was class imbalance. Draws make up only 26% of matches and are the hardest outcome to predict for any model. We addressed this with class-weighted loss functions and see improvement, but Draw prediction remains the weakest class — which is consistent with what professional betting markets also struggle with. We also had to be very careful about temporal feature computation: every rolling statistic must only look backwards in time. A single mistake here would contaminate results with future information.

[3:00 – 3:45] What Remains

We are very close to the finish line. The core technical work is complete and we are wrapping up the final details. What remains is the poster and the final presentation. We are also looking at some additional improvements and visual enhancements — better styled plots, an improved Streamlit interface, and possibly some extra analysis such as a per-league accuracy breakdown. We also have a working Streamlit web application where you can select any two teams, pick a date, and instantly see predicted probabilities for all three outcomes — which we plan to demonstrate live at the final presentation.

[3:45 – 4:00] Closing

In summary: the project is in a very strong position. Full pipeline complete, models trained and evaluated, GitHub repository live, Streamlit app working. We are doing very well and are excited to present the final result. Thank you — happy to take any questions.

7. Remaining Work

- Design and print the final A1 poster (Phase 7)
- Prepare the final 10-minute presentation slides
- Live demo of the Streamlit app during the final presentation
- Visual improvements to the project: enhanced plot styling, improved Streamlit UI layout, and additional charts such as a per-league accuracy breakdown and a head-to-head probability comparison view
- Optional: add SHAP explainability values to feature importance section